

University of Southern Denmark

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Take Home Exam 2024

Group 15

DM500 — Studyintroduction to Artificial Intelligence and Computer Science

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Contents

1	Introduction	3
2	Reexam february 2015	3
2.1	Task 1	3
2.2	Task 2	4
2.3	Task 3	5

1 Introduction

The objective of this assignment is the get to know **typst** as a tool for assignment writing. The following presents some exercises written in **typst**

2 Reexam february 2015

2.1 Task 1

Let $U = \{1, 2, 3, 4, \dots, 15\}$ be the universe Then let

$$A = \{2n \mid n \in S\} \quad B = \{3n + 2 \mid n \in S\}$$

Where $S = \{1, 2, 3, 4\}$

Declare all elements in the following sets

- a) A
- b) B
- c) $A \cap B$
- d) $A \cup B$
- e) $A - B$
- f) $\overline{A}$

2.1.1 Solution

- a) $\{2, 4, 6, 8\}$
- b) $\{5, 8, 11, 14\}$
- c) $\{8\}$
- d) $\{2, 4, 6, 5, 8, 11, 14\}$
- e) $\{2, 4, 6\}$
- f) $\overline{A} = U \setminus A = \{1, 3, 5, 7, 9, 10, 11, 12, 13, 14, 15\}$

2.2 Task 2

a) Which of the following statements are true?

1. $\forall x \in \mathbb{N} : \exists y \in \mathbb{N} : x < y$
2. $\forall x \in \mathbb{N} : \exists! y \in \mathbb{N} : x < y$
3. $\exists y \in \mathbb{N} : \forall x \in \mathbb{N} : x < y$

a) Negate proposition 1 from question a), without the use of $\neg$

2.2.1 Solution

- a) 1. $\forall x \in \mathbb{N} : \exists y \in \mathbb{N} : x < y \rightarrow \top$
2. $\forall x \in \mathbb{N} : \exists! y \in \mathbb{N} : x < y \rightarrow \perp$
3. $\exists y \in \mathbb{N} : \forall x \in \mathbb{N} : x < y \rightarrow \perp$

a) Negate proposition 1 from question a), without the use of $(\neg)$

$$\exists x \in \mathbb{N} : \forall y \in \mathbb{N} : x \geq y$$

2.3 Task 3

Let R, S and T be binary relations on the set $\{1, 2, 3, 4\}$

a) Let $R = \{(1, 1), (2, 1), (2, 2), (2, 4), (3, 1), (3, 3), (3, 4), (4, 1), (4, 4)\}$.

Is R a partial ordering?

b) Let $S = \{(1, 2), (2, 3), (2, 4), (4, 2)\}$.

Compute the transitive closure of S .

c) Let $T = \{(1, 1), (1, 3), (2, 2), (2, 4), (3, 1), (3, 3), (4, 2), (4, 4)\}$. Notice that T is an equivalence relation.

Compute T 's equivalence classes

2.3.1 Solution

a) R is reflexive since $(1, 1), (2, 2), (3, 3), (4, 4) \in R$.

R is antisymmetric: for every pair $(a, b) \in R$ with $a \neq b$, $(b, a) \notin R$ (checking $(2, 1), (2, 4), (3, 1), (3, 4), (4, 1)$, none of their reverses appear).

R is transitive: e.g. $(2, 4), (4, 1) \in R \rightarrow (2, 1) \in R$ and $(3, 4), (4, 1) \in R \rightarrow (3, 1) \in R$; all other compositions already lie in R .

Since R is reflexive, antisymmetric and transitive, R is a partial ordering.

b) Computing the reachability closure of $S = \{(1, 2), (2, 3), (2, 4), (4, 2)\}$:

$$S^+ = \{(1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (4, 2), (4, 3), (4, 4)\}$$

c) Grouping elements related by T :

1 and 3: $(1, 1), (1, 3), (3, 1), (3, 3) \in T$, so $\{1, 3\}$ is a class.

2 and 4: $(2, 2), (2, 4), (4, 2), (4, 4) \in T$, so $\{2, 4\}$ is a class.

Equivalence classes:

$$\{1, 3\} \quad \{2, 4\}$$

Task